General Mathematics — Exercise Sheet 17

Linear systems and elimination

Variant: Statements

Exercise 1 (A free column in the middle). Use Gauss–Jordan elimination to find every real solution of

$$\begin{aligned} x_1 + 2x_2 - x_3 + x_4 &= 1,\\ 2x_1 + 4x_2 + x_3 + 5x_4 &= 5,\\ -x_1 - 2x_2 + 2x_3 + 2x_4 &= 2. \end{aligned}$$

Identify the pivot columns and the free variable. Verify your family in the original equations.

Exercise 2 (A complex right-side parameter). For $\beta \in \mathbb{C}$, consider

$$(1+\mathrm{i})x_1 + x_2 = 2, \qquad 2x_1 + (1-\mathrm{i})x_2 = \beta, \qquad x_1, x_2 \in \mathbb{C}.$$

Find all β making the system consistent and give every solution in the consistent case. Use complex arithmetic from Lecture 15.

Exercise 3 (A parameter that can be zero). For every $a \in \mathbb{R}$, classify all real solutions of

$$ax_1 + x_2 = 1, \qquad x_1 + ax_2 = 1.$$

Do not lose the case $a = 0$ by dividing by the first coefficient. Give the exceptional parameter values and a full solution family whenever there are infinitely many solutions.

Exercise 4 (A proposed family and the actual free coordinate). A system has the reduced augmented matrix

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 0 & 3 \\ 0 & 1 & -1 & 0 & 4 \\ 0 & 0 & 0 & 1 & -2 \end{array}\right).$$

Find its pivot columns and every real solution. A proposed family is $x(t) = (3-2t, 4+t, t, t)$ for $t \in \mathbb{R}$. Find exactly which members of that proposed family solve the system, and explain which coordinate was incorrectly treated as free.

Exercise 5 (More unknowns and two independent parameters). For $c \in \mathbb{R}$, solve

$$x_1 + 2x_2 - x_3 = 1, \qquad 2x_1 + 4x_2 - 2x_3 = c.$$

Explain why having more unknowns than equations does not guarantee consistency. In the consistent case, give a complete family using separate parameters, and give a solution with $x_2 \neq x_3$. Explain why assigning the same parameter to both x_2 and x_3 would omit it.